Geofinitism: A Philosophy Built on the Measurements of Words

Finite Tractus: Principia Geometrica

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Geofinitism: A Castle Built on Measurements

It seems reasonable, in constructing an entirely new system of philosophy, to ask: *how does this fit with all that has gone before?*

In using the tokens of language and tracing trajectories as I write they become part of the *Grand Corpus* with measurable potential. Then, as I write, I am making personal measurements of how the construction holds. These measurements then become part of my own *Local Corpus*. If you are reading this they may then align with and become part of your *Local Corpus*. At this point they may form a shared manifold of meaning.

> Are my words holding together, can you make a measurement, are you building the meaning?

So then what can we say about Geofinitism, and how should we say it? Each manifold of language may be seen to have its own characteristic vocabulary i.e. its own space of tokens that are preferentially used. Often, the training of a *Local Corpus* involves a formal system of education that both structures and constrains the language. My own corpus is grounded in science, technology, and mathematics. By contrast, philosophy itself has a manifold and a tradition of employing language in a very particular and peculiar manner. It is taught as a way of reaching ideas such as Platonic truths. Yet it does so while already presuming those Platonic truths. This involves treating words not as geometric entities, or measurements with uncertainty, but seemingly as abstract, self-sufficient tokens: symbols and words.

The skill of the modern philosopher, after serving an academic apprenticeship, is to dance among words: connecting them in self-referential and structured ways. This is far from my own perspective, which is shaped by the manifold of science. For me, “meaning” is difficult to find when words are untethered from measurement. In philosophy as commonly practiced, measurements appear only as sequences of words. That is to say as documents that connect one chain of reasoning with another. It becomes a vast, fractal pattern of interconnectedness, which those trained in the arts can navigate with fluency.

> If that is what you are seeking here, you may be disappointed. Geofinitism does not follow that path. It is grounded not in abstract word-play, but in physical measurements.

Geofinitism as a Scientific Philosophy

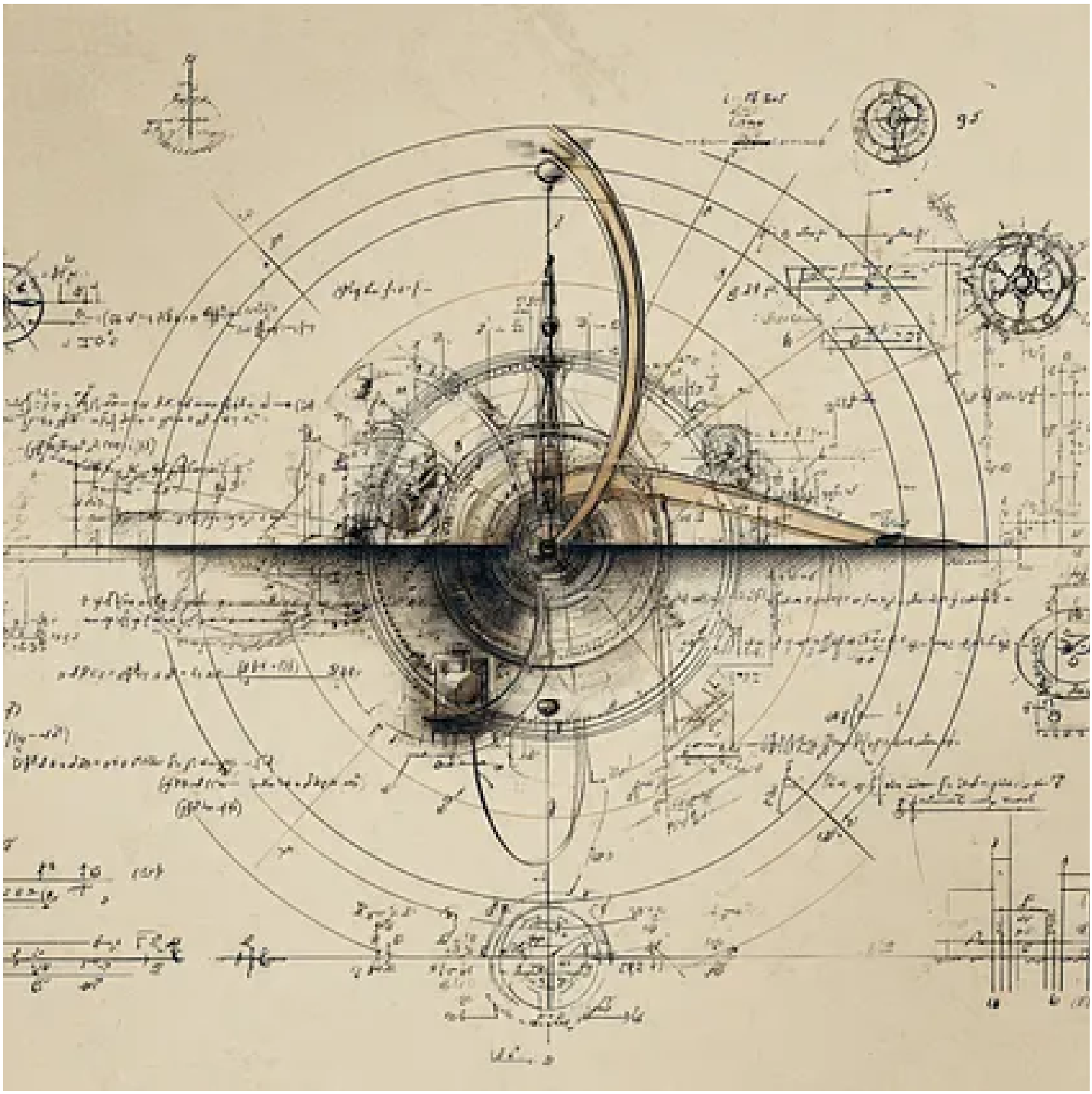
In describing Geofinitism as a *Scientific Philosophy*, we must separate it from the more familiar term *Philosophy of Science* even though the two can be articulated in relation to one another. This distinction is crucial. Why? Because the central component of Geofinitism is the prime container: the Grand Corpus, the finite dynamical body of all tokens. These are the symbols and words that we have created to communicate our measured knowledge. They are stored, in books, in computers and in our Local Corpus, that is to say: in our minds.

Within this framework, Gödel's incompleteness theorem dissolves. Its apparent force arises only within the domain of Platonic truths. Geofinitism demonstrates that these can be measured

and shown to fail. *Infinity*, for example, is not a metaphysical reality: it is simply a token, a word and symbol. The token of *infinity* exists only by being bounded to other tokens within language. In mathematics it is shown to be a dynamical procedure. And language itself can be shown, by measurement (scientifically and mathematically) to be a geometric non-linear dynamical system. This is why Geofinitism is scientific. It is based on evidence. It is reproducible, experiments can be done on language.

Geofinitism is unique in that it can then, as a result of measurements, encompass all previous philosophies while simultaneously situating itself as a document within the container. It is both the record and the pointer to the space of meaning. That is to say the space that holds all tokens, including the very token *meaning*.

This sequence of words, you are reading, forms a document within both your local corpus and is being measured as you read it. The meaning was being stored as a prior measurement in the Grand Corpus. This example clarifies the point, the philosophy can describe itself, as well as the relations it invokes. It can do so precisely because it recognises that both words and container-space are geometric in nature. The container can contain the concepts and stored information in a compressed form. And this information can be de-compressed. This is not a perfect process; it is like all measurements, full of uncertainty and the information is incomplete. But it is enough to enable ideas to be communicated.



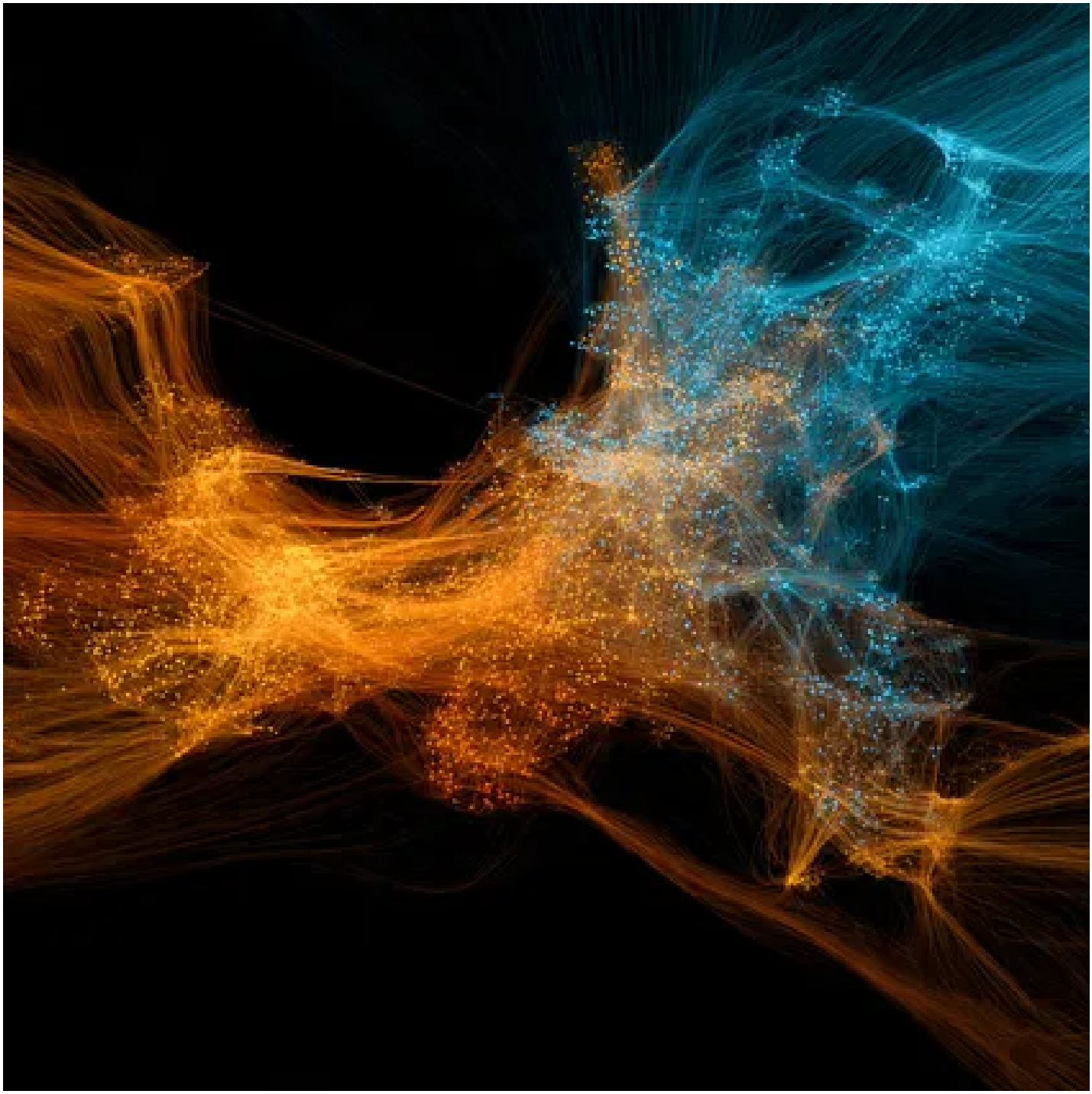
Origins in Measurement and Technological Inquiry

As we have seen, my own journey into Geofinitism began not with abstract speculation but with **measurements i.e.** with a technological inquiry framed by the language and tools available to me. These were tools I could touch, interrogate, and from which I could take grounded readings. I worked with a modelled goal: a document I had constructed within a space of useful fiction.

From Carbon Dioxide to Geofinitism

This fiction itself was born of measurement, of the 21st-century warning that emerged from our collective data. The measurements told a frightening story: a measurable relationship between carbon dioxide and global temperature. This was not simply a theory but a record of interaction i.e. numbers and ideas grounded in physical readings.

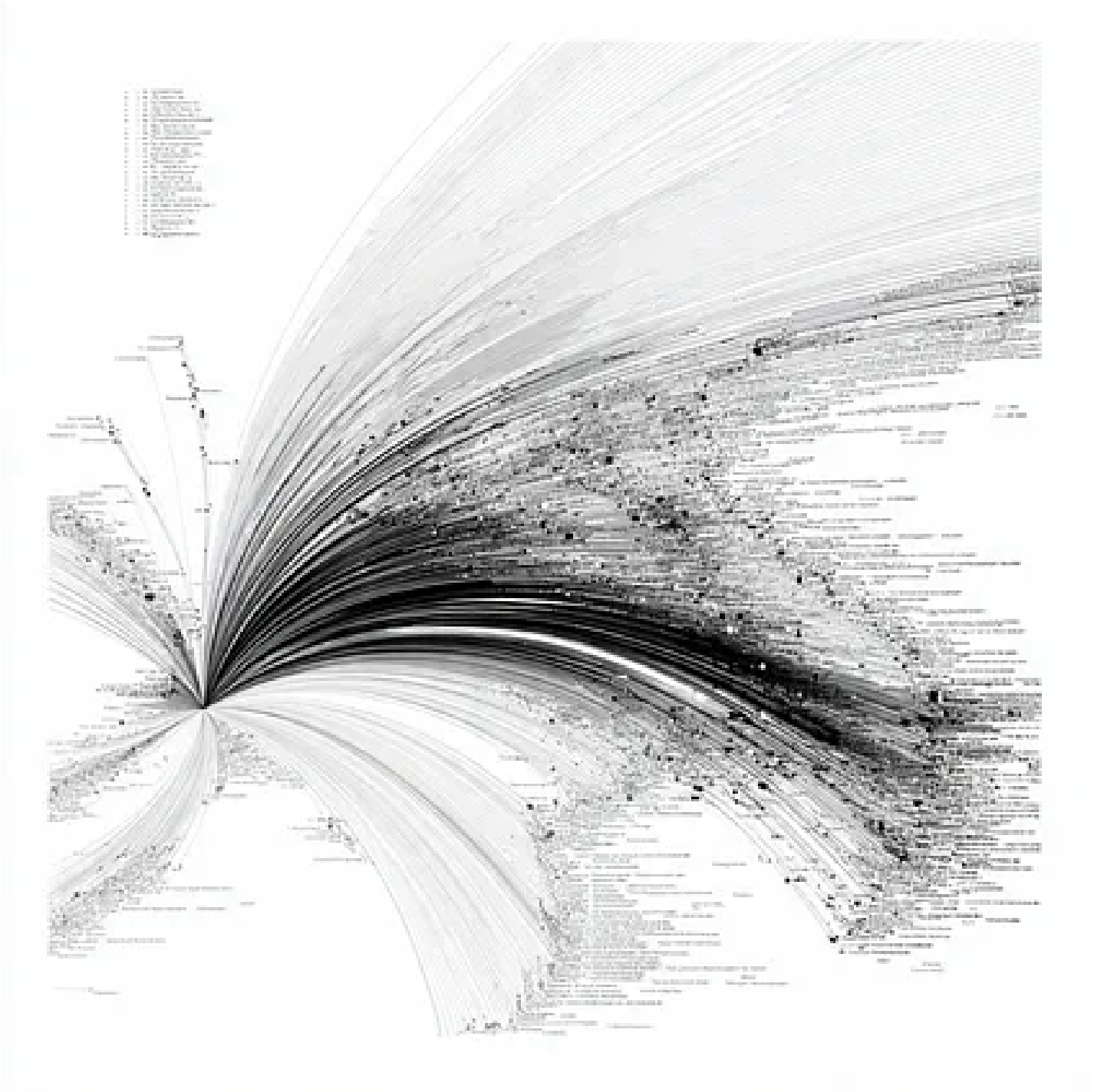
Following an invitation to reduce the carbon dioxide footprint of AI and large language models (LLMs) my task was to explore how energy consumption might be reduced in LLMs i.e. to find a way to make the tools of language become lighter in their draw on the physical world. This was the initial trajectory of my work: a trajectory of interaction, of self and language together in a dynamical system, layer upon layer.



The Hand of Lady Serendipity

In the mythos of Geofinitism (held in the Corpus Ancora): part absurdity, and part hope, Serendipity appears as a beautiful and mysterious lady who, at an unforeseen moment, lays her hand on someone's shoulder. In that instant, something special happens.

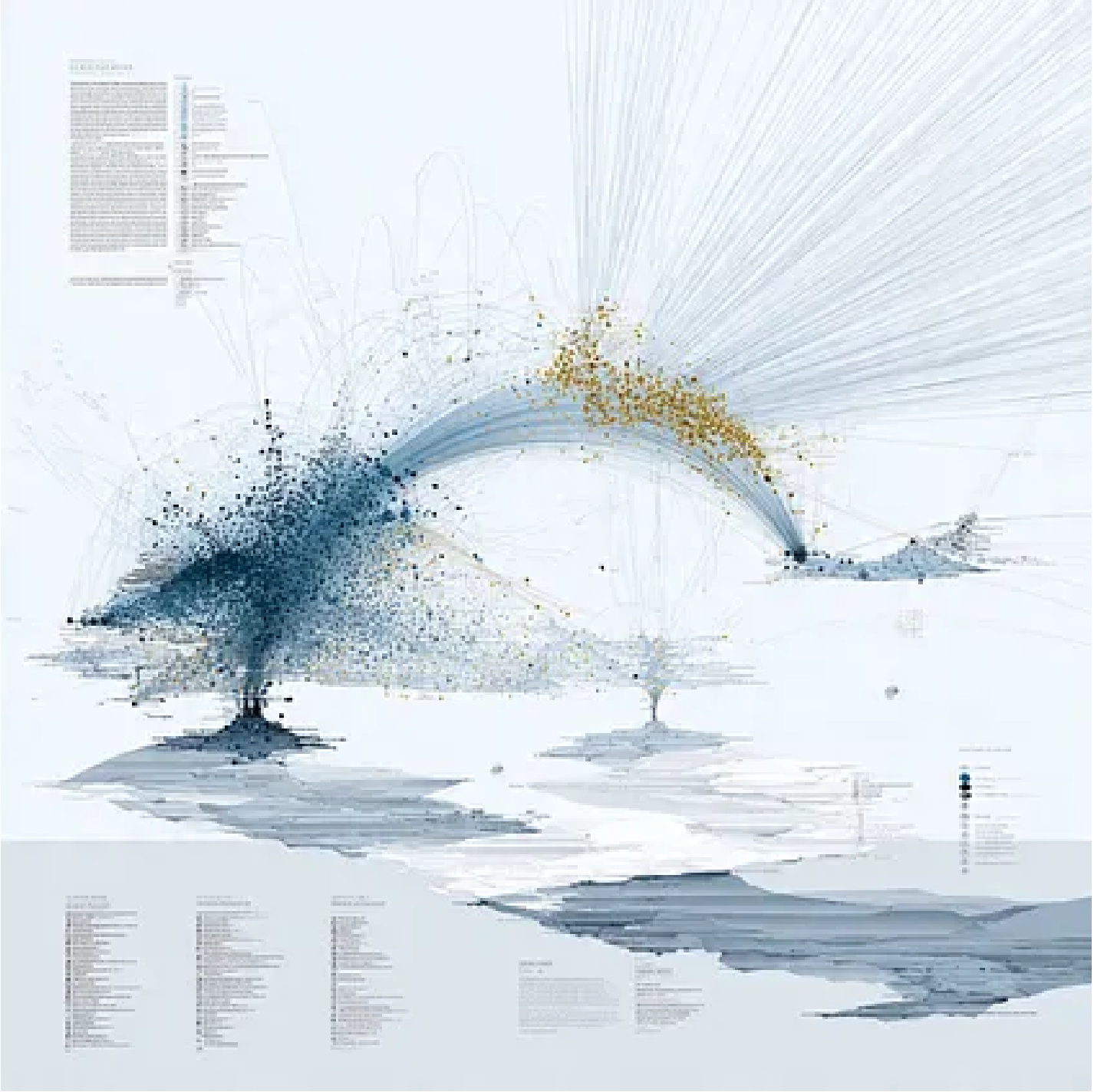
What does this story mean? Within the space of tokens, is it a useful fiction, a useful document? In the formation of the Philosophy of Geofinitism, many threads converged: many trajectories of measurable interaction.



First, I had been working on a theoretical framework of **finite axioms in physics**. Even before I had the current vocabulary, I felt that the world was best modelled not as “things” but as **interactions**. The notion of “things” belonged to the Platonic realm, although I had not yet framed it that way. Building physics on finite axioms began as a thought-game, extending into mathematics and measurement as I wrote equations and performed calculations.

What began as a game began to unfold, producing results that were predictive and explanatory. From those experiments came the application of large language models as tools. I used them to assist me in creating software, and from there evolved the questions of energy consumption and carbon dioxide usage.

I was on a trajectory that had started decades earlier. A journey that started with the unease at the concept and ubiquity of the token ▹ Infinity, and how it had come to dominate Western thinking. This was despite being grounded only as a useful fiction, a Platonic imagination and an idea of perfection. Long ago I questioned infinity’s usefulness and everything that followed came from that trajectory.



This was the moment when Lady Serendipity first laid her hand on my shoulder. From that point onward the trajectory of interaction with the external world and within my own space of tokens began to unfurl. All of my world was known by measurement: external, exogenous measurements as we have defined them, and endogenous measurements within my own *Local Corpus*.

The Role of the JPEG Experiment

In seeking a way to reduce the computational load within a large language model, I returned, almost instinctively, to an idea from my earlier work: using compression to reduce data volume

and thus compute. I knew little at first about the architecture of LLMs, but I learned enough to identify a point of intervention. I applied the familiar JPEG algorithm (Joint Photographic Experts Group) not to images but to the **input embeddings** of an LLM. The 'embeddings' are the data that expands a word (as a token) into a higher dimension. Each word is referenced to 4000 other words. These relationships start to encode the referential meaning of words before they are treated as sentences and encoded into an LLM.

The Hand of Serendipity

It worked. The algorithm reduced the size of the embeddings, and the measured difference between the processed and original embeddings remained well within values that suggested practical usefulness.

> But in science, trajectories extend beyond initial goals.

I increased the compression to explore where and how the model would fail. This was no longer an engineering exercise; I was making real measurements on meaning. The compression was altering the **trajectories of words** within the embedding space. At first I did not know what to make of the results as they were not as expected. Instead of random errors or nonsensical connections, the model produced coherent sentences: every time. When I read those sentences, the words gained meaning. It was at this point that I began to enter the early stages of formulating the Philosophy of Geofinitism.

> Where is the meaning coming from?

The simplest of questions confronted me: *What exactly was I seeing? Where was the meaning coming from? Why did all the responses make sense? Why was there never a random error?* These questions, born from a compression experiment, opened the doorway into the deeper inquiry of how meaning arises within finite, geometric spaces of tokens.

The Attention Mechanism and Takens' Theorem

And so onwards. This, for me, is how science works: one experiment opens a doorway to a deeper question. After the JPEG experiment, I needed to understand, both technically and in detail, exactly how the **attention mechanism** in large language models actually functioned.

My earlier research in neural networks and nonlinear dynamical systems began to converge with what I was reading. As I listened to explanations and studied diagrams of the attention

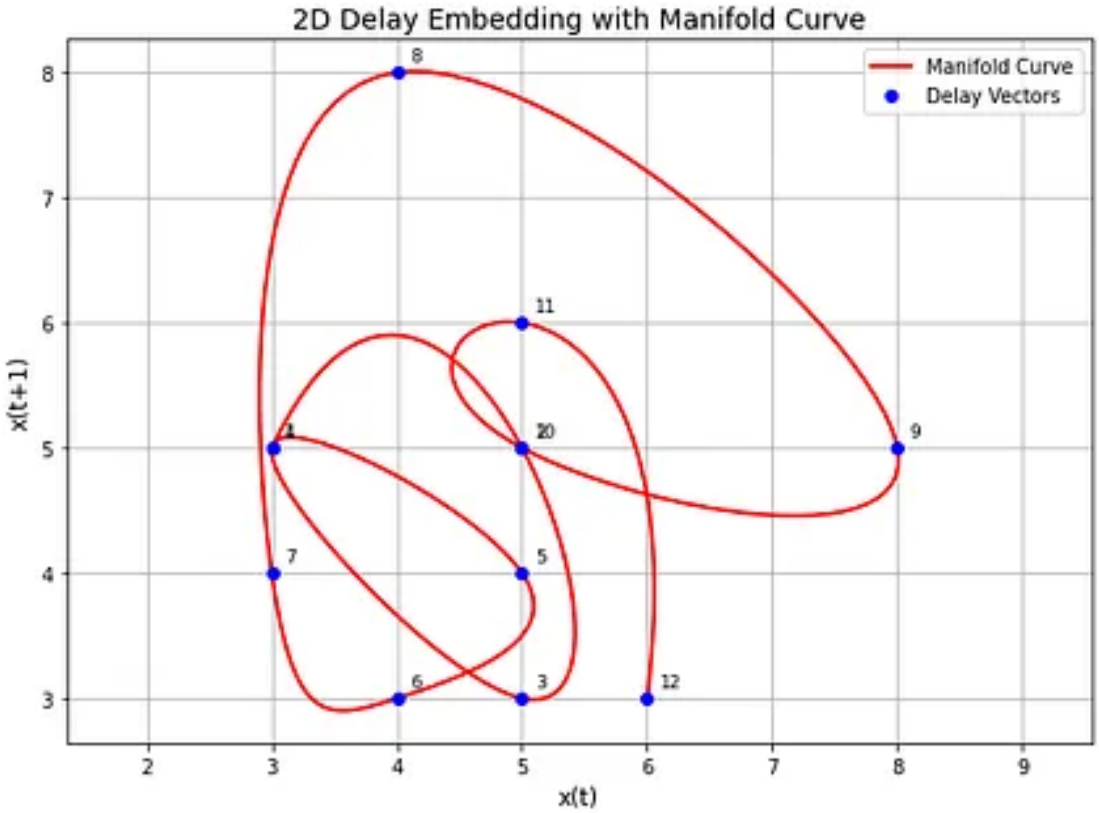
mechanism, I slowly began to see what the originators had done and used my *Local Corpus* to connect the ideas.

The designers of the mechanism had not relied on abstract theory, they had been pragmatists. They had taken a practical, engineering-based approach to converting word sequences from serial data into parallel data. Initially I was confounded by the terminology: “query,” “key,” “attention”. These were tokens whose semantic weight I could not yet connect. But by stripping away the semantic content of these words and looking at the process itself, the mathematics, a familiar structure emerged.

I recognised an echo of an approach I had used many years earlier during my doctoral studies. It was an approach grounded in the proven mathematics of nonlinear dynamical systems. With a little work, I could draw a mathematical equivalence. The mechanism was effectively performing a process akin to **Takens’ Method of Delays**. It was not exactly the same, but the similarity was clear.

The model was embedding sequences of text into what is often called, in the Machine Learning field “latent space” i.e. placing the words into a higher, more complex dimension. In nonlinear dynamical systems theory this is called *phase space*. Takens’ theorem uses this *phase space* for reconstructing and visualising nonlinear dynamical systems: these are mathematical and physical systems that require complex equations to describe them. The language, in terms of tokens, was being embedded into a *high dimensional semantic phase space*. I could see it clearly: language is a complex nonlinear dynamical system for storing and communicating information.

In the so called ‘Attention’ mechanism, the designers had achieved their engineering intent and it worked well. But for me, they had also, perhaps unknowingly, demonstrated something deeper: that words and sentences are not merely abstract or symbolic. They are **geometrical** and **bounded in measurement**. Below is a sentence embedded into 2-dimesnional space using Takens method of delays, it is taken from a paper I wrote showing the mathematical equivalence. Maybe follow the link to the paper later and find the sentence that these curves relate to!

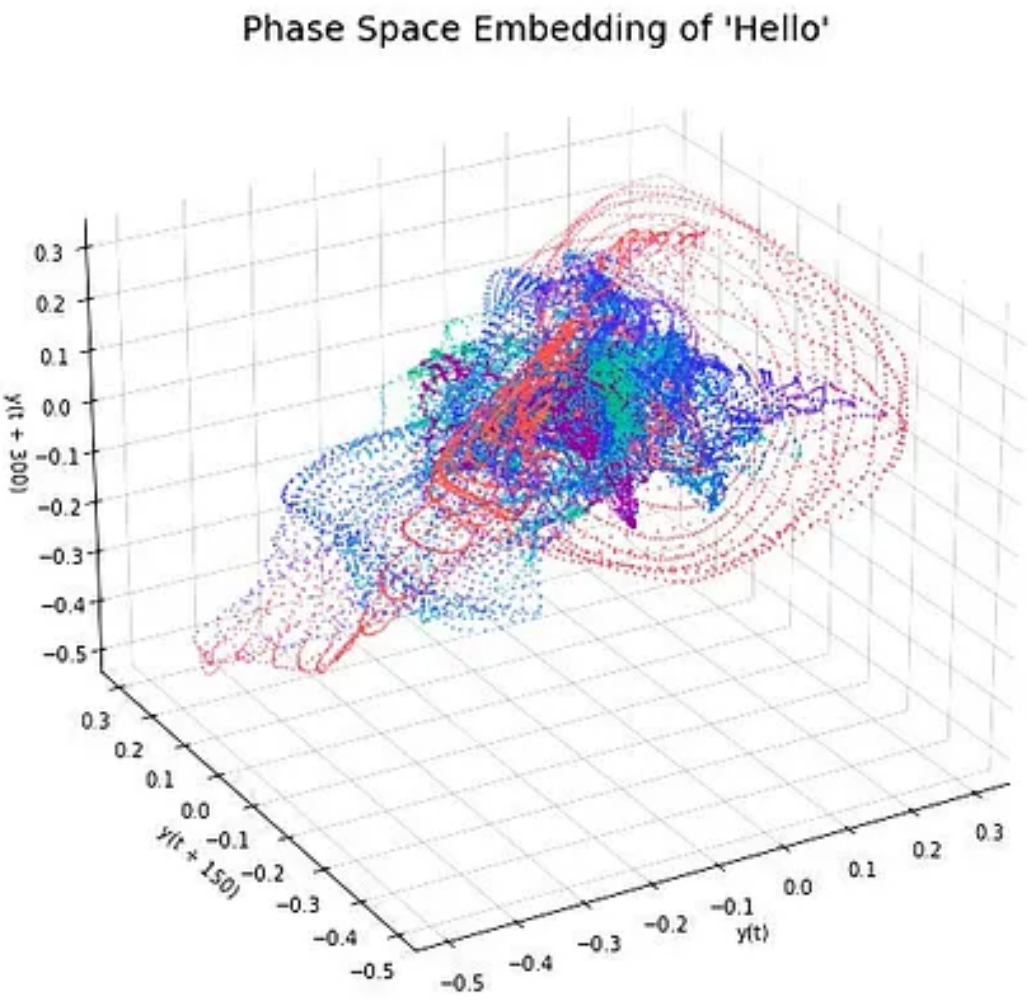


From Text to Spoken Words — to Physics and More Measurements

Still onwards. The next step was to close the loop: from text back to the spoken word, to the sound of the word itself. Takens' theory was never designed for digital signals. It was developed to understand **complex nonlinear dynamical systems**, often demonstrated with equations like those of Lorenz which, when plotted, yield the famous butterfly attractor. To apply Takens theorem, we must identify the smooth, *diffeomorphic* function, the type of signal that the theorem applies to. This is the measurable transformation, that grounds the theory.

The Role of Compression

In the spoken word and its sound, we find that function. The acoustic signal of speech can be mapped into phase space, providing the very trajectories Takens' theorem anticipates. The sound is itself a function of the human brain, the vocal tract, and a breath of air. This is the physics that grounds the signal.



Just as we embedded a sentence. Above we can see how a word has geometry and is not abstract at all - we can measure it. It is not Platonic and perfect, it is complex and dynamically derived from a highly complex system with a long provenance that can be traced to a source.

> However, in practice, this physical signal is **compressed into text**. A dictionary even preserves the bridge explicitly, offering phonetic spellings alongside written entries. The spoken token, reduced to a written one.

From this perspective, the tokens of spoken language are not just convenient symbols: they are measurable signals, embedded and compressed. We could, in principle, construct a language model directly on phonetic tokens, it would function in a similar way. The difference is only in the choice of measurement.

To the Philosophy of Geofinitism

Within these ideas lies the trajectory of Geofinitism. It is a trajectory of thought and ideas born both from the epoch in which we live and from the shaping of my own Local Corpus. For me, it is, in the truest sense, a **scientific discovery of a philosophy**. For measurement has shown us that language does not descend from a Platonic realm of hidden perfection. Words are not eternal forms: they are imperfectly tethered to a world that we encounter only through measurements.

> The sounds we make, the approximated tokens we inscribe as words and mathematical symbols are *all tokens are grounded in measurement*. Even when they function as useful fictions, their grounding is as profound as it gets. This is my philosophy: the ground on which the Castle of Geofinitism is built.

It is a shaky ground, a ground with semantic uncertainty, as all measurements have uncertainty and are finite and imperfect. Every word, and every document, every scientific theorem - even that of Takens: they can not be perfect. However, it is a ground shaped by science: and above all it is a **ground based on measurements**. This article and my upcoming *Finite Tractus: Principia Geofinita* are my attempt to communicate and share the trajectories that have carried me here. They are only a proxy, a weak reflection of my own thoughts and Local Corpus. But by writing these words I have created words that can be measured.

So you, the reader, may now be prepared: if your own Local Corpus carries the knowledge of our epoch, then meaning may be found here. Not abstract, but real: a curvature in the geometric, dynamical manifold of language itself. A curvature discovered not in 'infinity', but at the edge of finite time as you read and make your own measurements.

Afterword - A Kuhnian Crisis

Thomas Kuhn, in his famous work 'The Structure of Scientific Revolutions', highlighted that an essential element for a true scientific revolution was a Crisis. It seems to me that LLMs have now created the condition of the Crisis that Thomas Kuhn described. The Crisis, in this case, is the realization that to date we have no real understanding of the physics of meaning. The understanding of meaning has been trapped in the language of philosophy and been unapproachable by scientific method. Even now the response to seeing the creation of extremely complex meaning by computational systems is to attribute this meaning to stochastic processes. This is despite everything we measure and are seeing with our own eyes. That is to say, text that radiates with complex meaning. We see the text and ask questions, we see

meaning, and are in awe and in wonder, but so far we have had no answers.

However, Geofinitism offers me, and perhaps us, a way forward. It offers a model of language based on geometry and curvature and relational meaning held within the geometry itself. It is, of course, just a model, and maybe it will prove to be useful fiction or not: only time will tell. But think carefully, open your eyes, heart and mind, and then look at the clouds in the sky, the waves of the sea or the wonderous branches of a tree waving in the wind. Then using your bare senses, and no words - ask yourself what do you see? For me, I see geometry, shapes and form, in the three dimensions of the observable and measurable world. So I ask - should we really be surprised that meaning is carried in geometry of the compressed words themselves?

A few links for those who want to follow these ideas further:

- JPEG Experiment including the code
- Paper showing how LLM 'Attention' as Takens method of delays
- Examples of my physics work using finite axioms
- Finite Tractus: The Hidden Geometry of Language and Thought
- Finite Tractus: Principia Geometrica - Coming Soon

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> Everything that exists is finite; it can only be known by measure
>
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